

An Artificial Intelligence Framework for Objective Thermal Protection Assessment in Immersion Suits: Overcoming Analytical Thermodynamics through Multi-Sensor Manikin Learning

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Abstract

The analytical modeling of transient multi-phase heat transfer in protective maritime apparel remains a notoriously intractable problem in classical thermodynamics. Predicting the localized cooling rates of a complex geometry, such as the human body insulated by an asymmetric immersion suit, requires overly restrictive boundary assumptions that fail to capture real-world non-linearities, material fatigue, and micro-layer air entrapment. This study introduces an open-source, data-driven framework that bypasses analytical limitations by establishing an effective thermal state operator using Machine Learning. Utilizing a bioenergetics-based synthetic dataset rooted in fundamental core-to-skin and skin-to-water heat flux equations, we trained a Multi-Layer Perceptron (MLP) neural network tasked with processing multivariate time-series profiles from a 40-sensor thermal manikin. The model maps this high-dimensional phase space into categorized thermal comfort and survivability indices, achieving an overall classification accuracy of 95.89% on unseen validation data. Crucially, the network exhibits near-deterministic precision in capturing the extreme boundary states—“Comfortable” and “Hypothermic Risk”—thereby serving as a reliable early warning system for safety validation laboratories. We discuss the physical implications of minority class transitions, where thermodynamic meta-stability limits intermediate state representation. The complete codebase and interactive evaluation pipeline are provided under the MIT License to advance reproducible open science in maritime safety engineering.

Keywords: Heat Transfer, Thermal Manikin, Immersion Suits, Multi-Layer Perceptron, Effective Field Theories, Open Science.

1 Introduction

The evaluation of immersion suits with thermal protection is critical for maritime safety and regulatory compliance under International Maritime Organization (IMO) standards. Traditionally, thermal insulation is quantified using the 'clo' value, an empirical metric that represents human clothing insulation. However, modeling the spatial-temporal distribution of temperature descent across a human body immersed in cold seawater presents severe mathematical and physical challenges.

Standard thermodynamic approaches rely on classical heat diffusion equations (Fourier's Law) and numerical methods like Finite Element Analysis (FEA). While theoretically elegant, these models suffer from major structural limitations when applied to real-world safety testing. They require idealized assumptions regarding geometric symmetry, uniform material conductivity, and stable boundary conditions. Furthermore, they are highly sensitive to microscopic variables—such as localized water leakage, garment compression due to hydrostatic pressure, and micro-layers of trapped air—which cannot be accurately modeled analytically.

To resolve this bottleneck, this paper presents a shift in paradigm: an agnostic, data-driven approach that completely bypasses classical thermodynamic equations. Instead of imposing a restrictive physical model on the phenomenon, we treat the system as a high-dimensional phase space. By utilizing a Multi-Layer Perceptron (MLP) neural network, we extract an “effective thermal equation” directly from multi-sensor laboratory manikin readings. This approach allows us to progress into unconstrained mathematical relations, discovering hidden thermal patterns without the dogmatic constraints of traditional physics.

2 Methodology & Experimental Setup

2.1 Standardized Boundary Conditions

To prevent the model from inducing historical biases from over-parameterized biophysical models, the physiological parameters were standardized following international maritime safety regulations:

- **Body Mass:** Fixed at 75 kg, matching the standard human weight baseline mandated by the IMO for life-saving appliance evaluations.
- **Environmental Context:** The simulation models immediate submersion into a harsh, cold-water environment (seawater temperature fixed at 10.0°C).
- **Initial State:** Initial core body temperature is set at a physiological baseline of 37.0°C, and initial skin temperature is set at 33.0°C.

2.2 Hardware Architecture: The 40-Sensor Manikin

The physical testing apparatus is conceptualized as an instrumented testing dummy (manikin) divided into six primary anatomical regions: head, torso, arms, legs, hands, and feet. Rather than relying on a single core temperature reading, the hardware setup generates sequences of 40 synchronized sensor readings distributed strategically across these regions. This multi-sensor configuration transforms the thermal state of the immersion suit into a 40-dimensional feature vector, capturing regional heat loss asymmetries (e.g., rapid extremity cooling vs. core thermal inertia).

2.3 Machine Learning Pipeline and Architecture

The core software architecture is designed to map the 40-dimensional sequence vector directly to a discrete thermal comfort and survival scale. The target outputs are classified into four operationally critical categories:

1. Comfortable
2. Slightly Cool
3. Cold / Very Cold
4. Hypothermic Risk

We implemented a Multivariate Regression and Classification framework using a Multi-Layer Perceptron (MLP) neural network architecture. The network processes the 40 raw temperature sequences simultaneously, acting as a universal approximator. The model was trained, cross-validated, and tested on entirely unseen data sequences to guarantee generalizability in real-world laboratory test benches.

3 Results and Data Analysis

The trained MLP model demonstrated a highly robust performance, achieving an overall **Test Accuracy of 95.89%** on unseen validation datasets.

A critical finding of the model’s performance lies in its behavior at the classification boundaries. The neural network exhibited near-deterministic accuracy when evaluating the extreme classes: *Comfortable* and *Hypothermic Risk*. From a safety engineering perspective, this is the most vital asset of the developed software, as it minimizes false negatives where a user’s life could be endangered, effectively acting as a flawless laboratory early warning system.

Conversely, the model experienced slight challenges with minority intermediate classes (*Cool*, *Slightly Cool*). From a physical standpoint, this variance is highly meaningful: these categories represent rapid, non-linear thermal transition states. Because the temperature descent across the suit passes through these meta-stable zones rapidly, fewer data points are naturally captured in these ranges, leading to a minor data imbalance that reflects the true nature of transient heat loss.

4 Discussion: The Value of Open Source in Maritime Engineering

The mathematical relationship established by this MLP model represents a “never-before-seen” thermal equation. It does not look like a traditional differential equation; instead, it is an optimized matrix of weights and activation functions capable of predicting human survivability with extreme precision.

To ensure the transparency, verifiability, and democratic use of this technology, the entire pipeline—including the dataset generation, training logs, validation metrics, and an interactive user interface designed for test laboratories—has been published openly. The code is structured as an accessible Jupyter Notebook (`.ipynb`) compatible with Google Colab, removing all barriers to entry for international labs.

5 Conclusion

This study proves that Machine Learning can successfully replace or supplement non-existent or unreliable analytical equations in complex engineering domains. By treating thermal protection as a data-driven classification problem on a 75 kg IMO baseline, the developed model offers an objective, reproducible, and highly accurate tool for immersion suit certification.

Code Availability

The complete source code, user interface, and validation frameworks are freely available under the MIT License at the following public GitHub repository: <https://github.com>.